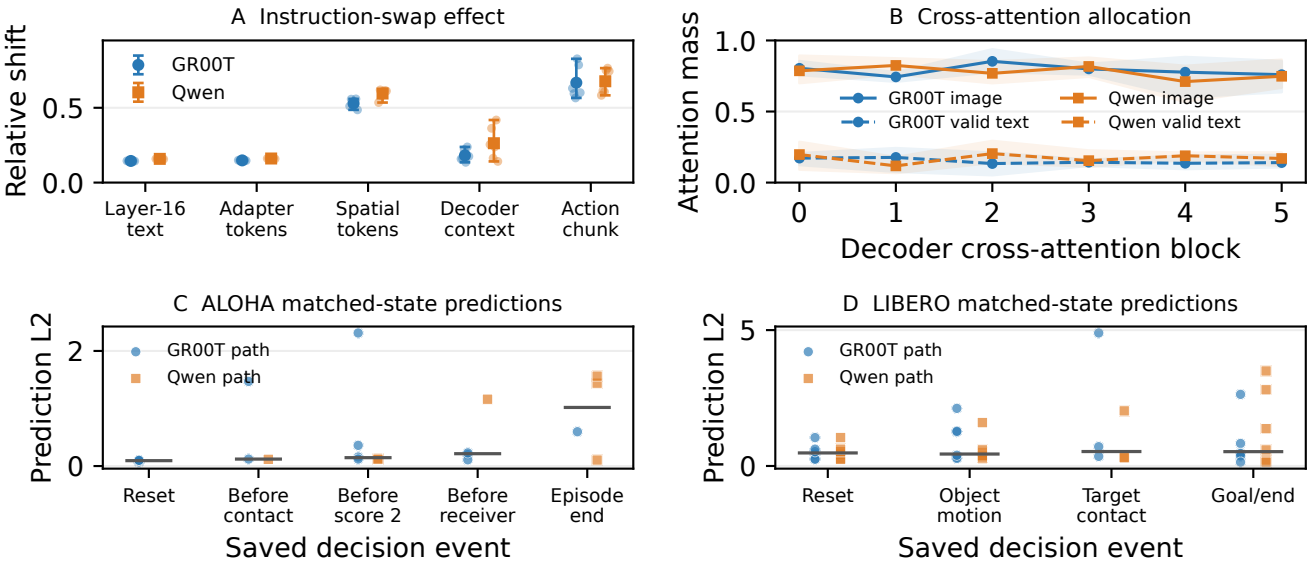
A Instruction-swap effect
Relative shift
GR00T
Qwen
0.0
0.5
Layer-16 text
Adapter tokens
Spatial tokens
Decoder context
Action chunk
B Cross-attention allocation
Attention mass
1.0
0.5
0.0
GR00T image
GR00T valid text
Qwen image
Qwen valid text
0
1
2
3
4
5
Decoder cross-attention block
C ALOHA matched-state predictions
Prediction L2
GR00T path
Qwen path
2
0
Reset
Before contact
Before score 2
Before receiver
Episode end
Saved decision event
D LIBERO matched-state predictions
Prediction L2
GR00T path
Qwen path
5
0
Reset
Object motion
Target contact
Goal/end
Saved decision event